

Derivation: ZS and The Birthday Paradox

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1. The Birthday Paradox: From 365 Days to Generalization

The classic Birthday Paradox asks for the probability that at least two people in a room share a birthday, assuming a year has 365 days. To compute this, it is easier to calculate the complement: the probability that *no one* shares a birthday ($P(A')$), and subtract it from 1.

For k people and 365 days:

- The 1st person has 365 available days: $\frac{365}{365}$
- The 2nd person has 364 available days: $\frac{364}{365}$
- The k -th person has $365 - k + 1$ available days: $\frac{365-k+1}{365}$

The probability of all unique birthdays is the product of these fractions:

$$P(A') = \frac{365}{365} \times \frac{364}{365} \times \cdots \times \frac{365 - k + 1}{365} = \frac{P(365, k)}{365^k}$$

Generalizing to $N = 2^n$ days: If we replace 365 with $N = 2^n$ days, the formula for k people having unique birthdays becomes:

$$P(A') = \frac{N \times (N - 1) \times \cdots \times (N - k + 1)}{N^k} = \frac{P(2^n, k)}{(2^n)^k} = \frac{P(2^n, k)}{2^{nk}}$$

2. The Probability Formula & Common Denominator

Let $P(A)$ be the probability that at least two people share a birthday. Using the complement rule:

$$P(A) = 1 - P(A')$$

Substituting our generalized formula:

$$P(A) = 1 - \frac{P(2^n, k)}{2^{nk}}$$

To express this as a single fraction $\frac{A_{\text{raw}}}{B_{\text{raw}}}$, we union the denominators by rewriting 1 as $\frac{2^{nk}}{2^{nk}}$:

$$P(A) = \frac{2^{nk}}{2^{nk}} - \frac{P(2^n, k)}{2^{nk}} = \frac{2^{nk} - P(2^n, k)}{2^{nk}}$$

Thus, before simplification, our numerator and denominator are:

$$\begin{aligned} A_{\text{raw}} &= 2^{nk} - P(2^n, k) \\ B_{\text{raw}} &= 2^{nk} \end{aligned}$$

3. The Pigeonhole Principle

Before calculating large numbers, we must handle the edge case where a shared birthday is an absolute certainty. If the number of people (k) is strictly greater than the number of available days (2^n), then by the Pigeonhole Principle, at least two people *must* share a birthday.

$$\text{If } k > 2^n \implies P(A) = 1 \implies \text{Output } 1/1 \text{ (or } 1 \text{)}$$

4. Simplifying the Fraction: GCD Calculation

We are required to output the irreducible fraction $\frac{A}{B}$, which means dividing A_{raw} and B_{raw} by their Greatest Common Divisor (GCD).

Looking at the denominator $B_{\text{raw}} = 2^{nk}$, we see it is a pure power of 2. It has no other prime factors. Therefore, the GCD of A_{raw} and B_{raw} is simply the highest power of 2 that divides A_{raw} .

Let E be the number of times 2 divides A_{raw} . Because 2^{nk} is astronomically larger than $P(2^n, k)$, the trailing zeros of the difference $2^{nk} - P(2^n, k)$ are dictated entirely by the trailing zeros of the smaller term, $P(2^n, k)$.

$$E = \nu_2(P(2^n, k)) = \nu_2(2^n \times (2^n - 1) \times \cdots \times (2^n - k + 1))$$

Let us analyze the factors of 2 in each term:

- The first term is 2^n , contributing exactly n factors of 2.
- For any subsequent term $(2^n - i)$ where $i < 2^n$, the number of trailing zeros is identical to the trailing zeros of i . This is because subtracting from a higher power of 2 does not alter the lowest set bits.

Thus, the remaining $k - 1$ terms contribute the exact same number of 2s as the product $1 \times 2 \times 3 \cdots \times (k - 1) = (k - 1)!$.

$$E = n + \nu_2((k - 1)!)$$

By Legendre's Formula, the highest power of 2 dividing $X!$ is $X - \text{popcount}(X)$. Substituting $X = k - 1$:

$$E = n + k - 1 - \text{popcount}(k - 1)$$

Our simplified equations before modulo are now:

$$\begin{aligned} A &= (2^{nk} - P(2^n, k)) / 2^E \\ B &= 2^{nk} / 2^E = 2^{nk-E} \end{aligned}$$

5. When $V = 0$ (The Subtrahend)

We must compute the answers modulo $M = 10^6 + 3$. Let $V = P(2^n, k) \pmod{M}$ be the term we subtract from 2^{nk} .

The permutation $P(2^n, k)$ is the product of k consecutive descending integers. If $k \geq M$, the sequence of numbers being multiplied is guaranteed to contain at least one multiple of M . Any number multiplied by a multiple of M becomes 0 $\pmod{M}$.

$$\text{If } k \geq 10^6 + 3 \implies V \equiv 0 \pmod{M}$$

If $k < M$, we can safely iterate and compute V in $O(k)$ time.

6. Overcoming nk Overflow with Euler's Totient Theorem

We need to calculate $2^{nk} \pmod{M}$. However, the constraints dictate $n, k \leq 10^{18}$. This means the exponent $n \times k$ can reach up to 10^{36} , which massively overflows a 64-bit unsigned integer (`uint64_t` max is $\approx 1.8 \times 10^{19}$).

To avoid this overflow, we use Euler's Totient Theorem, which states $a^{\varphi(M)} \equiv 1 \pmod{M}$. Here, $\varphi(M)$ is Euler's Totient function, representing the number of positive integers up to M that are coprime to M . Since our modulo $M = 10^6 + 3$ is a prime number, $\varphi(M) = M - 1$. This specific case is known as Fermat's Little Theorem:

$$a^{\varphi(M)} \equiv 1 \pmod{M}$$

This implies that we can reduce the exponent modulo $\varphi(M)$ before doing the exponentiation:

$$2^{nk} \equiv 2^{(nk \pmod{\varphi(M)})} \pmod{M}$$

To compute $(nk \pmod{\varphi(M)})$ without overflow, we apply the modulo to n and k individually before multiplying:

$$\text{Exponent} = (n \pmod{\varphi(M)}) \times (k \pmod{\varphi(M)}) \pmod{\varphi(M)}$$

This securely fits inside standard integer types, allowing us to compute $B \equiv 2^{\text{Exponent}} \pmod{M}$ in $O(\log M)$ time using binary exponentiation.